PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T15 Classified

Cross-session • chapter-organised candidate

11 classified items • 37 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Unit 2 • Chapter OL-T15 • 11 items • 37 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Direct & Inverse Proportion

Unit 2 • 37 marks • 11 item(s)

June 2025 | 4MA1-1H Q18 | 4 marks

Recover the input under inverse variation

June 2024 | 4MA1-2H Q17 | 3 marks

Root direct variation y=ksqrtx

June 2023 | 4MA1-2H Q16 | 5 marks

Root direct variation y=ksqrtx

November 2024 | 4MA1-1H Q14 | 3 marks

Inverse-square or inverse-power variation

June 2022 | 4MA1-2H Q17 | 4 marks

Power direct variation y=kx^n

January 2022 | 4MA1-1H Q15 | 3 marks

Recover the input under inverse variation

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

May-Nov 2020 | 1H Q14 | 3 marks

Inverse-square or inverse-power variation

January 2023 | 1H Q17 | 3 marks

Inverse-square or inverse-power variation

November 2025 | 2H Q17 | 3 marks

Inverse-square or inverse-power variation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

18 y is inversely proportional to x^3

$y = 4$ when $x = 3$

Work out the value of x when $y = 864$

$x =$ ....

(Total for Question 18 is 4 marks)

Worked Solution 11

Inverse proportionality

UNIT 2
4 marks

Find the constant

$$y = \frac{k}{x^3}, \quad 4 = \frac{k}{3^3} \Rightarrow k = 108.$$

Use y=864

$$864 = \frac{108}{x^3} \Rightarrow x^3 = \frac{1}{8} \Rightarrow x = \frac{1}{2}.$$

Final answer

$$x = \frac{1}{2}$$

Question 22

4MA1-2H Q17

ORIGINAL QUESTION
3 marks

17 Q is directly proportional to the square root of d

$Q = 4.5$ when $d = 324$

Find a formula for Q in terms of d

....

Worked Solution 22

Direct proportion

MODULAR UNIT 2

3 marks

Constant

$$Q = k\sqrt{d}, \quad 4.5 = 18k \Rightarrow k = 1/4.$$

Final answer

$$Q = \frac{\sqrt{d}}{4}$$

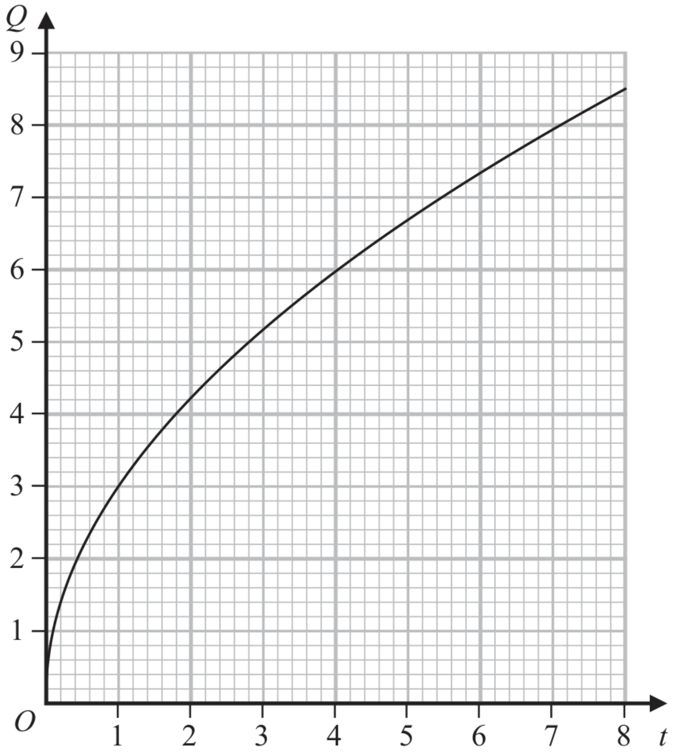
Question 26

4MA1-2H Q16

ORIGINAL QUESTION
5 marks

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

.... %
(2)

Worked Solution 26

Root proportionality and percentage change

MODULAR UNIT 2
5 marks

Read the graph

$$Q \propto \sqrt{t}, \quad Q = k\sqrt{t};\ \text{a point such as } (1,3) \text{ gives } k = 3.$$

Write the model

$$Q = 3\sqrt{t}.$$

Apply the 20% change

$$Q' = 1.2Q \Rightarrow 3\sqrt{t'} = 1.2(3\sqrt{t}) \Rightarrow t' = 1.2^2 t = 1.44t.$$

Convert to a percentage

$$(1.44 - 1) \times 100\% = 44\%.$$

Final answer

$$Q = 3\sqrt{t}; \quad 44\%$$

Original question 11

4MA1-1H Q14 M01

MODULAR UNIT 2
3 marks

14 B is inversely proportional to the square of d

$B = 0.25$ when $d = 12$

Find a formula for B in terms of d

....

Worked solution 11

Model inverse-square variation

MODULAR UNIT 2
3 marks

M01 – Model inverse-square variation

Write the model

$$B = k/d^2.$$

Use the given pair

$$0.25 = k/12^2 \Rightarrow k = 36.$$

State the formula

$$B = 36/d^2.$$

Final answer

$$B = \frac{36}{d^2}$$

Original question 29

4MA1-2H Q17 Q17

ORIGINAL QUESTION

4 marks

17 M varies directly as the cube of h

$M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

....

Worked solution 29

Chapter: Direct & Inverse Proportion

MODULAR UNIT 2
4 marks

Q17 – Chapter: Direct & Inverse Proportion

Family: Build an algebraic direct-variation model • Variant: Power direct variation y=kx^n

Method
Direct variation gives M = kh^3.
Why: Translate the variation statement into an equation.

Method
4 = k(0.5)^3, so k = 32.
Why: Use the given pair to determine the constant.

Method
500 = 32h^3, so h^3 = 15.625.
Why: Substitute the new M value and isolate h cubed.

Method
h = cube root of 15.625 = 2.5.
Why: Take the real cube root.

Final answer
h: h = 2.5

Original question 10

4MA1-1H Q15

ORIGINAL QUESTION
3 marks

15 A is inversely proportional to C^2

$A = 40$ when $C = 1.5$

Calculate the value of C when $A = 1000$

$C =$ ....

Worked solution 10

Chapter: Direct & Inverse Proportion

MODULAR UNIT 2
3 marks

Q15 – Chapter: Direct & Inverse Proportion

Family: Build an inverse-variation model • Variant: Recover the input under inverse variation

Method
A=k/C^2 and 40=k/(1.5)^2, so k=90.
Why: Use inverse proportionality to find the constant.

Method
When A=1000, 1000=90/C^2, so C^2=0.09.
Why: Substitute the new value of A.

Method
C=0.3 (taking the positive value for this quantity).
Why: Take the positive square root.

Final answer
C: 0.3

16 A is inversely proportional to the square of r

$A = 5$ when $r = 0.3$

(a) Find a formula for A in terms of r

....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$ ....
(3)

Worked solution

June 2021 | Question 16 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method / working

1. Write inverse-square proportionality as $A=k/r^2$.
2. Use $A=5$ when $r=.3$: $5=k/(.3)^2$, so $k=.45$.
3. Therefore $A=.45/r^2$.

Final answer: $A=0.45/r^2$

16 A is inversely proportional to the square of r

$A = 5$ when $r = 0.3$

(a) Find a formula for A in terms of r

....

(3)

(b) Find the value of A when $r = 7.5A$

$A =$ ....

(3)

Worked solution

June 2021 | Question 16 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method / working

1. Substitute $r=7.5A$ into $A=.45/r^2$.
2. Rearrange: $A^3=.45/(7.5^2)=.008$.
3. Take the positive cube root: $A=.2$.

Final answer: $A=0.2$

14 F is inversely proportional to the square of v.

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v.

....

(Total for Question 14 is 3 marks)

Answer reference | May-Nov 2020 | Question 14

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

14	$F=\frac{k}{v^2}$ **or** $Fv^2=k$ oe		3	M1	(NB. Not for $F=\frac{1}{v^2}$) Constant of proportionality must be a symbol such as k	M2 for $6.5=\frac{k}{4^2}$ oe
	$6.5=\frac{k}{4^2}$ **or** $k=6.5\times4^2$ **or** $k=104$			M1	For substitution of F and v into a correct formula	
		$F=\frac{104}{v^2}$		A1	Award 3 marks if $F=\frac{k}{v^2}$ is on the answer line and the value of $k=104$ is found	
						Total 3 marks

Worked solution

May-Nov 2020 | Question 14 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method

F is inversely proportional to v^2, so

$F=\frac{k}{v^2}$

Use $F=6.5$ when $v=4$:

$6.5=\frac{k}{4^2}$

$k=6.5 \times 16=104$

Therefore

$F=\frac{104}{v^2}$

Answer: $F=\frac{104}{v^2}$.

January 2023 | 1H | Question 17 | 3 marks

Chapter OL-T15: Direct & Inverse Proportion

17 y is inversely proportional to $\sqrt{x}$

$y = c^4$ when $x = c^2$ where c is a positive constant.

Find a formula for y in terms of x and c
Give your answer in its simplest form.

....

(Total for Question 17 is 3 marks)

Answer reference | January 2023 | Question 17

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

17	$y=\frac{k}{\sqrt{x}}$ or $ky=\frac{1}{\sqrt{x}}$ or $\sqrt{x}=\frac{k}{y}$ oe		3	M1 (NB. Not for $y=\frac{1}{\sqrt{x}}$) Constant of proportionality must be a symbol such as k (Allow c for k for this mark only)	M2 for $c^4=\frac{k}{\sqrt{c^2}}$ oe
	$c^4=\frac{k}{\sqrt{c^2}}$ oe or $k=c^4\times\sqrt{c^2}$ oe			M1 for substitution of x and y into a correct formula	
	Correct answer scores full marks (unless from obvious incorrect working)	$y=\frac{c^5}{\sqrt{x}}$		A1 oe e.g $y=c^5\times\frac{1}{\sqrt{x}}$ Award 3 marks if answer is $y=\frac{k}{\sqrt{x}}$ on the answer line and $k=c^5$ clearly given in the body of working of the script	
				Total 3 marks	

Worked solution

January 2023 | Question 17 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Official final mark-scheme pages are embedded immediately after this question.

17 T is inversely proportional to $\sqrt{m}$

$T = 15$ when $m = 36$

Find a formula for T in terms of m

....

(Total for Question 17 is 3 marks)

Worked solution

November 2025 | Question 17 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method
T is inversely proportional to m, so
$T=\frac{k}{m}$
Use $T=15$ when $m=36$:
$15=\frac{k}{36}$
$k=540$
Therefore
$T=\frac{540}{m}$
Answer: $T=\frac{540}{m}$.

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION